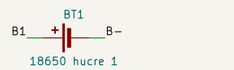
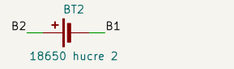
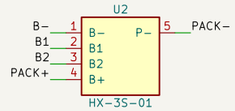
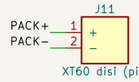
PIL, BMS, SARJ TEMASI

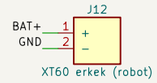


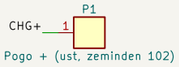










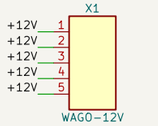


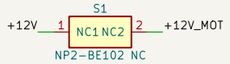




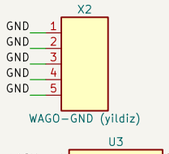
12 V DAGITIM

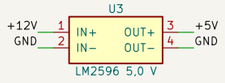




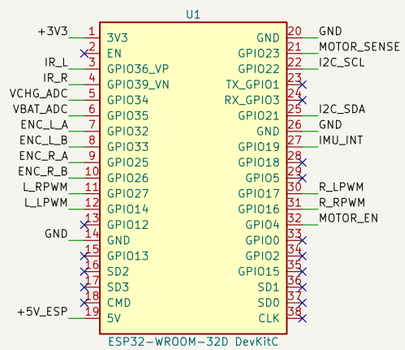






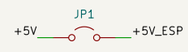


ANA KART: ESP32 VE BESLEME



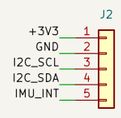


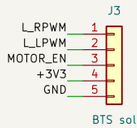


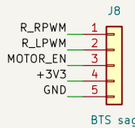


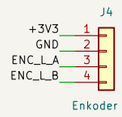


ANA KART: KONNEKTOR, OLCUM

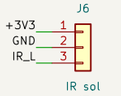


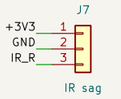


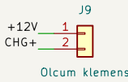


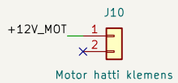




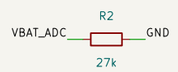




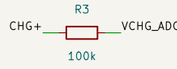






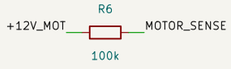




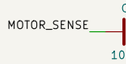




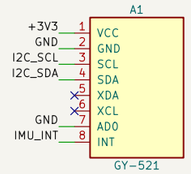


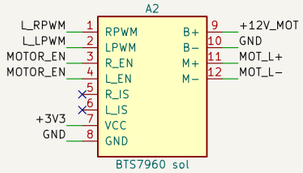


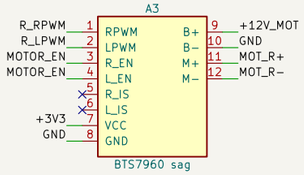


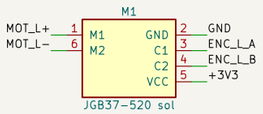


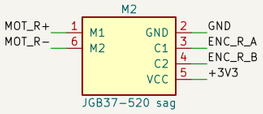
MODULLER VE MOTORLAR

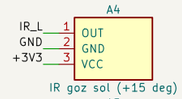


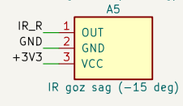












CDTP denge robotu — robot elektronigi (tek sayfa). Ayni adli etiketler ayni hattir.

Guc: pil -> BMS -> XT60 -> F1 5 A -> WAGO-12V. Acil stop (NC) yalniz motor hattini keser; ESP32 acik kalir.
Sarj: pogo + -> D1 SR5100 -> BAT+ (sigortadan once). Robotta acikta pil gerilimi yok.

JP1: kod yuklerken (USB takiliyken) cikarin; LM2596 ile USB 5 V'u cakismasin.
BTS7960 VCC = 3,3 V (ESP32 seviyesi). R_EN ve L_EN iki kartta GPIO4'e; R5 10k ile acilista motorlar kapali.

MPU6050 ADO -> GND (adres 0x68). GPIO36/39 yalniz giris (IR, PCNT 12,8 us suzgec).
Bolucu: R1/R2 ve R3/R4 100k/27k (12,6 V -> 2,68 V); R6/R7 100k/33k (12,6 V -> 3,13 V).

Kablo numaralari (W1-W44) ve pin sirasi: docs/robot.html bolum 04, 06, 07.

Uretici: tools/kicad_sema.py (elle duzenlemeyin; betigi degistirip yeniden uretin)

YTU EEF CDTP

Sheet: /
File: cdtp_robot.kicad_sch

Title: CDTP denge robotu — robot elektronigi

Size: A3	Date: 2026-10-29	**Rev: v1.3**
KiCad E.D.A. 10.0.6		Id: 1/1